2026 Level 2 - Derivatives

Pricing and Valuation of Forward Commitments

a. describe the carry arbitrage model without underlying cashflows and with underlying cashflows

b. describe how equity forwards and futures are priced, and calculate and interpret their no-arbitrage value

c. describe how interest rate forwards and futures are priced, and calculate and interpret their no-arbitrage value

d. describe how fixed-income forwards and futures are priced, and calculate and interpret their no-arbitrage value

e. describe how interest rate swaps are priced, and calculate and interpret their no-arbitrage value

f. describe how currency swaps are priced, and calculate and interpret their no-arbitrage value

g. describe how equity swaps are priced, and calculate and interpret their no-arbitrage value

LOSs will match between the video and the PDFs, but may be in a different order than the CFAI readings

Forwards/Futures Prices

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⇒ Arbitrage-free pricing & valuation

assumptions/
- replicating instruments are identifiable and investable
- market frictions are nil
- short selling is allowed with full use of proceeds
- borrowing & lending are available at a known risk-free rate

Notation: S - underlying
F - forward | V - value of forward
f - futures | v - value of futures

$V_0 = 0$
- at contract initiation, the value of a futures/forward contract = 0 (i.e. no money changes hands)
- at expiration, $F_T = f_T = S_T$ (called convergence)

$V_T = F_T - F_0 = S_T - F_0$ (long) $\quad V_T = F_0 - F_T = F_0 - S_T$ (short)

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- futures/ $v_t = f_t - f_{t-1}$ (before)
$v_t = 0$ (after)
- futures are marked-to-market daily

1/ no underlying cash flows $F_0 = S_0e^{rT}$ - cont. comp.
or/ $F_0 = S_0(1 + r)^T$ - periodic comp.

e.g./ $S_0 = 100$
$r_f = 5\%$
$T = 1$ yr.

Step #1. Borrow \$100 for one year
Step #2. Buy S_0
Step #3. Sell F_0
Step #4. Payback loan of S_0e^{rT} at time T

$F_0 = S_0e^{rT} = 100e^{.05(1)} = 105.127$
$F_0 = S_0(1 + r)^T = 100(1.05)^1 = 105$

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e.g./ $S_0 = 100$
$r_f = 5\%$
$T = 1$ yr.

$F_0 = S_0 e^{rT} = 100e^{.05(1)} = 105.127$
or/ $F_0 = S_0(1 + r)^T = 100(1.05)^1 = 105$

general rule : buy low, sell high

Case 1: $F_0 = 110$
- sell F_0 (i.e. short)
- borrow \$100, buy S_0
- at T, deliver S_0 for \$110
- pay back \$105
- at time 0, borrow $5/1.05 = 4.762$ (get paid today)

carry arbitrage

Case 2: $F_0 = 90$
- Sell S_0 for \$100, invest @ 5%
- Buy F_0 (i.e. long)
- at T, take delivery of S for \$90, cover short position
- profit = \$15
- at time 0, borrow $15/1.05 = 14.286$ (get paid today)

reverse carry arbitrage

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An Australian stock paying no dividends is trading in Australian dollars for A\$63.31, and the annual Australian interest rate is 2.75% with annual compounding. Based on the current stock price and the no-arbitrage approach, what is the equilibrium three-month forward price?

$S_0 = 63.31$
$r_f = 2.75\%$
$T = .25$

$F_0 = S_0(1 + r)^T$
$= 63.31(1.0275)^{.25} = 63.74$

If the interest rate immediately falls 50 bps to 2.25%, the three-month forward price will: ↓

$63.31(1.0225)^{.25} = 63.66$

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1/ no underlying cash flows Key point: the quoted forward price does <u>not</u> directly reflect expectations of future underlying prices

⇒ Value

long

0 t T

short

S_t

$F_0 = S_0(1+r)^T$

$F_t = S_t(1+r)^{T-t}$

$$V_t = \frac{F_t - F_0}{(1+r)^{T-t}} \quad \text{or/} \quad V_t = S_t - F_0/(1+r)^{T-t}$$

e.g./ $F_0 = 105$
$T = 1$ yr.
$S_t = 110$
$t = 9$ mos.
$r_f = 5\%$

$V_0 = 0$ $\quad F_0 = 105$

0 $\quad S_t = 110 \quad$ T

$$F_t = S_t(1+r)^{T-t} = 110(1.05)^{.25} = 111.3499$$

$$110 - 105/(1.05)^{.25} = 6.2729$$

$$V_t = \frac{(111.3499 - 105)}{(1.05)^{.25}} = 6.2729$$

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2/ Underlying with cash flows

γ – gamma ⇒ benefits
θ – theta ⇒ costs

$$F_0 = [S_0 + PV(\theta) - PV(\gamma)](1+r)^T$$

increase F_0 $\quad$ decrease F_0

e.g./ $S_0 = 100$
$r_f = 5\%$
$T = 1$
CF = 2.9277 @ t = .5

$$F_0 = \left(100 + 0 - \frac{2.9277}{(1.05)^{.5}}\right)(1.05)^1 = 102$$

- 3 mos. later $S_t = 105$

$$F_t = \left(105 + 0 - \frac{2.9277}{(1.05)^{.25}}\right)(1.05)^{.75} = 105.9134$$

$$V_t = (105.9134 - 102)/(1.05)^{.75} = 3.7728$$

$$\left(105 - \frac{2.9277}{(1.05)^{.25}}\right) - 102/(1.05)^{.75} = 3.7728$$

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2/ Underlying with known yield

$\underbrace{\text{known yield}}_{\text{assumed to be continuous}}$

$$F_0 = S_0 e^{(r_c + \theta - \gamma)T}$$

$\left.\begin{matrix} r_c \\ \theta \\ \gamma \end{matrix}\right\}$ all continuous rates

Recall/ $(1+r)^T = e^{r_c T} \Rightarrow \ln[(1+r)^T] = r_c T$

$$\frac{T \ln(1+r)}{T} = r_c$$

$$r_c = \ln(1+r)$$

e.g./ r_f = 5% annual

$r_c = \ln(1.05) = 4.897\%$ & $e^{.04897} = 1.05$

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A. Equities/

The continuously compounded dividend yield on the EURO STOXX 50 is 3%, and the current stock index level is 3,500. The continuously compounded annual interest rate is 0.15%. Based on the carry arbitrage model, the three-month futures price will be *closest* to:

$\gamma = 3\%$ $r_c = .15\%$ $F_0 = S_0 e^{(r_c - \gamma)T}$

$S_0 = 3500$ $T = .25$ $= 3500e^{(.0015-.03).25} = 3475.15$

Suppose Nestlé common stock is trading for CHF70 and pays a CHF2.20 dividend in one month. Further, assume the Swiss one-month risk-free rate is 1.0%, quoted on an annual compounding basis. Assume that the stock goes ex-dividend the same day the single stock forward contract expires. Thus, the single stock forward contract expires in one month.

$$F_0 = (S_0 - PV(\gamma))(1+r)^T$$

$$F_0 = S_0(1+r)^T - \gamma = 70(1.01)^{1/12} - 2.20 = 67,858$$

$$(S_0 - PV(\gamma))(1.01)^{1/12}$$

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Suppose we bought a one-year forward contract at 102 and there are now three months to expiration. The underlying is currently trading for 110, and interest rates are 5% on an annual compounding basis. If there are no other carry cash flows, the forward value of the existing contract will be closest to:

0 — 9 (t) — T

$F_0 = 102$

$S_t = 110$

$$F_t = 110(1.05)^{.25} = 111.3499$$

$$\frac{\begin{array}{r} 111.3499 \\ -102 \end{array}}{9.3499}$$

$$110 - 102/(1.05)^{.25} = 9.2365$$

$$9.3499/(1.05)^{.25} = 9.2365$$

If a dividend payment is announced between the forward's valuation and expiration dates, assuming the news announcement does not change the current underlying price, the forward value will most likely:

$\downarrow$ $[S_0 + PV(\theta) = PV(\gamma)]$ **decrease**

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B. Interest Rates

- forward rate agreement (OTC)
  - underlying is an interest rate on a deposit

    long = floating rate receiver (pays fixed)
    short = fixed rate receiver (pays floating)

  - no exchange of notional amount
  - FRA price is the fixed interest rate that eliminates arbitrage

    3x9 FRA – the rate on a deposit that begins in 3 months and lasts for 6 months

L(90)
L(270)

0 — 3 — 9

f(3,6) = FRA fixed rate

$\left(30/360\right)$

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B. Interest Rates

L(90) f(3,3)
L(180)
FRA expires FRA payoff
discounted back

advanced set, advanced settled (FRAs)

- advanced set, settled in arrears
- swaps
- interest rate options

- receive floating

$$\frac{\text{NA}(\text{floating rate} - \text{fixed rate})\left(\text{days}/360\right)}{\left[1 + \text{floating rate}\left(\text{days}/360\right)\right]}$$

- receive fixed

$$\frac{\text{NA}(\text{fixed rate} - \text{floating rate})\left(\text{days}/360\right)}{\left[1 + \text{floating rate}\left(\text{days}/360\right)\right]}$$

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B. Interest Rates

In 30 days, a UK company expects to make a bank deposit of £10,000,000 for a period of 90 days at 90-day Libor set 30 days from today. The company is concerned about a possible decrease in interest rates. Its financial adviser suggests that it negotiates today, at Time 0, a 1 × 4 FRA, an instrument that expires in 30 days and is based on 90-day Libor. The company enters into a £10,000,000 notional amount 1 × 4 receive-fixed FRA that is advanced set, advanced settled. The appropriate discount rate for the FRA settlement cash flows is 0.40%. After 30 days, 90-day Libor in British pounds is 0.55%.

rec. fx. = fl.

L(30) f(1,3)
0 30 120
L(120) L(90) = .55%

The interest actually paid at maturity on the UK company's bank deposit will be closest to:

$$10\text{M}\left(.0055\left(90/360\right)\right) = 13{,}750$$

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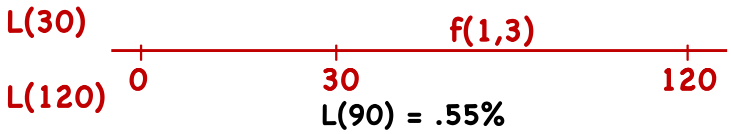
B. Interest Rates



If the FRA was initially priced at 0.60%, the payment received to settle it will be closest to:

rec. fx. - pay fl.

$$\frac{\text{NA}(\text{fx. rate} - \text{fl. rate})\left({}^{90}/_{360}\right)}{1+.004\left({}^{90}/_{360}\right)} = \frac{10\text{M}(.006-.0055)\left({}^{90}/_{360}\right)}{1+.004\left({}^{90}/_{360}\right)}$$

$$= 1248.75$$

If the FRA was initially priced at 0.50%, the payment received to settle it will be closest to:

$$\frac{10\text{M}(.005-.0055)\left({}^{90}/_{360}\right)}{1+.004\left({}^{90}/_{360}\right)} = \frac{-1250}{1.001} = -1248.75$$

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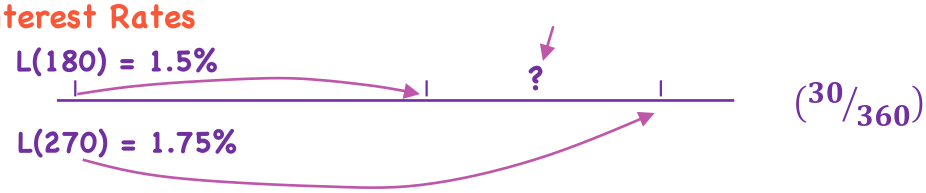
B. Interest Rates



6 x 9 FRA fixed rate = ?

$$\left[1+.0175\left({}^{270}/_{360}\right)\right] = \left[1+.015\left({}^{180}/_{360}\right)\right]\left[1+x\left({}^{90}/_{360}\right)\right]$$

$$\frac{1+.0175\left({}^{270}/_{360}\right)}{1+.015\left({}^{180}/_{360}\right)} = 1+x\left({}^{90}/_{360}\right)$$

$$1.005583127 = 1+x\left({}^{90}/_{360}\right)$$

$$x = 2.233\%$$

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B. Interest Rates

Suppose we entered a receive-floating 6 x 9 FRA at a rate of 0.86%, with notional amount of C$10,000,000 at Time 0. The six-month spot Canadian dollar (C$) Libor was 0.628%, and the nine-month C$ Libor was 0.712%. Also, assume the 6 x 9 FRA rate is quoted in the market at 0.86%. After 90 days have passed, the three-month C$ Libor is 1.25% and the six-month C$ Libor is 1.35%.

Assuming the appropriate discount rate is C$ Libor, the value of the original receive-floating 6 x 9 FRA will be closes to:

NA(fl. - fx.)

L(180) = .628%

L(270) = .712%

.86%

x

0 T

6x9

L(90) = 1.25%

L(180) = 1.35%

$$\left[\frac{1+.0135\left(\frac{180}{360}\right)}{1+.0125\left(\frac{90}{360}\right)}-1\right]\times 4$$

$$=.01445$$

(rec. fl., pay fx.) (pay fl., rec fx.)

-.86% +1.45%

$$\frac{10M(.0145-.0086)\,\frac{90}{360}}{1+.0135\left(\frac{180}{360}\right)}=14,651.10$$

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C. Fixed Income Forwards

1. Full price = Clean Price + Accrued Interest
   (quoted price)
2. Fixed Income futures contracts are based on a generic bond i.e. Treasury Bond Futures – one contract involves the delivery of $100,000 face value, 6% semi-annual coupon

∴ underlying ≠ deliverable – short position decides what to deliver
e.g./ T-Bond futures ⇒ any gov't. bond > 15 yrs. to maturity as of Day 1 of the delivery period

∴ need a conversion factor – quoted price a bond would have on the first day of the delivery period on the assumption that the interest rate for all maturities = 6%

$\underbrace{\text{(Quoted futures price}}_{\text{T-bond quote}} \times \text{CF}) + \text{AI}$ } defines the price to be paid to the short side on delivery

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C. Fixed Income Forwards

3. Cheapest-to-Deliver (CTD) – since the short side decides which bond to deliver, they will choose the CTD

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C. Fixed Income Forwards

borrow

I

0 T

buy bond S_0

• pay AI $\underline{AI}$

$\left[S_0 + AI - {}^{I}/_{(1+r)^t}\right](1+r)^T$ —— • owe $\left[(S_0 + AI) - {}^{I}/_{(1+r)^t}\right](1+r)^T$

• sell S_T = ?

• get AI (known amt.)

$$F_0 = \left[(S_0 + AI) - {}^{I}/_{(1+r)^t}\right](1+r)^T - AI$$

$$F_0 = QF_0 \times CF$$

$$QF_0 = F_0/CF$$

C. Fixed Income Forwards

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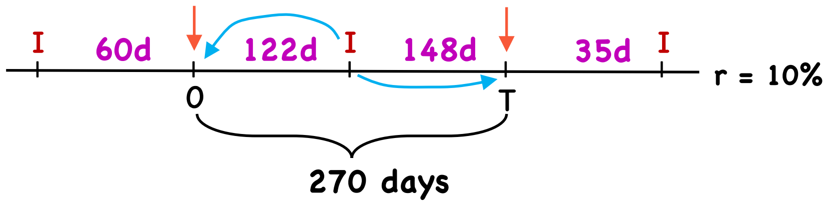

CTD
12%
PV = 120
CF = 1.4

$$F_0 = \left[(S_0 + AI) - {}^{I}/_{(1+r)^t}\right](1+r)^T - AI$$

$$\left[\left(120 + 6\left({}^{60}/_{182}\right)\right) - {}^{6}/_{(1.1)^{122/365}}\right](1.1)^{270/365} = 124.651885$$

$$\frac{-\,6\left({}^{148}/_{183}\right)}{119.799426}$$

$$QF_0 = {}^{F_0}/_{CF}$$

$$= {}^{119.799426}/_{1.4} = 85.571$$

C. Fixed Income Forwards

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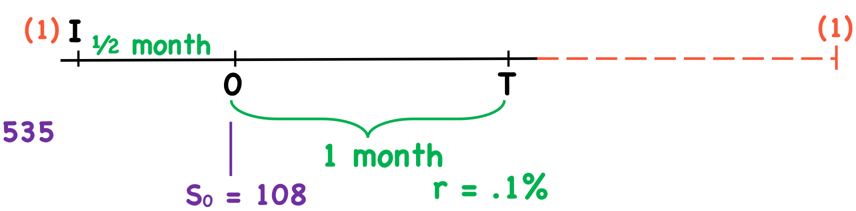
CTD
(2% semi)
CF = .729535



S_0 = 108
AI = 0.083

$$\left(QF_0 = {}^{F_0}/_{CF}\right) \qquad QF_0 = \frac{\left[(S_0 + AI) - {}^{I}/_{(1+r)^T}\right](1+r)^T - AI}{CF}$$

$$= \frac{[(108 + .083) - 0](1.001)^{1/12} - 1\left({}^{1.5}/_{6}\right)}{.729535}$$

$$= {}^{107.842}/_{.729535} = 147.82$$

$$F_0 = QF_0 \times CF = 147.82 \times .729535 = 107.842$$

C. Fixed Income Forwards

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1 month

0 t T r = .1%

2 months

long 5 forwards
F_0 = 145
(underlying = $100,000)

F_t = 148

F_0 = 145
F_t = 148
(148 − 145)

$$V_t = \frac{(148 - 145)}{(1.001)^{1/12}}$$

$$= 2.9997$$ /100 of par value

$$\text{Total Value}_t = 5 \times \frac{100{,}000}{100} \times 2.9997 = 14{,}998.50$$

Swaps

A. Interest Rate Swaps/

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Company A ⟵ fixed ⟵ Dealer ⟵ fixed ⟵ Company B

Company A ⟶ Libor ⟶ Dealer ⟶ Libor ⟶ Company B

swap rate

bid
- the fixed rate a dealer is willing to pay for Libor

spread • spread

ask
- the fixed rate a dealer is willing to accept for Libor

swap rate

rate at which V_{swap} = 0 @ T = 0

Company A/ receive fixed, pay floating

equivalent to:
- long a fixed rate bond
- short a floating rate bond

the swap rate is the fixed rate such that
$B_{fx} = B_{fl}$

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A. Interest Rate Swaps/

- since B_{fl} = 100 at a reset date, solve for PMT to find the fixed rate

$$1 = \frac{PMT}{(1+r_1)} + \frac{PMT}{(1+r_2)^2} + \cdots + \frac{PMT+1}{(1+r_n)^n}$$

• notional = $1

zero rates for maturities t_1, t_2 … t_n

Since $^1/_{(1+r)^t} = DF_t$

$$1 = PMT \cdot DF_1 + PMT \cdot DF_2 + \cdots + PMT \cdot DF_n + DF_n$$

$$1 = PMT\left(\sum_{i=1}^{n} DF\right) + DF_n$$

$$PMT \cdot \sum DF = 1 - DF_n$$

$$PMT = \frac{1 - DF_n}{\sum DF}$$

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A. Interest Rate Swaps/

e.g./ 5 yr. Libor-based interest rate swap with annual resets

Maturity	DF
1	.99099
2	.977876
3	.965136
4	.951529
5	.937467
	4.822107

swap rate:

$$r_{fix} = \frac{1 - DF_n}{\sum DF} = \frac{1 - .937467}{4.822107}$$

$$= .012968$$

$$\sim 1.2968\%$$

Valuation: $V_{swap} = (r_{fix_0} - r_{fix_t})\sum DF \bullet NA$

rec. fx., pay fl. T=0
rec fl., pay fx. T=*t*

(given current Libor)

$$V_{swap} = V_{fix_0} - V_{fix_t}$$

$$= (PMT_0 \cdot DF_1 + PMT_0 \cdot DF_2 + \cdots + PMT_0 \cdot DF_n + DF_n)$$

$$= (PMT_t \cdot DF_1 + PMT_t \cdot DF_2 + \cdots + PMT_t \cdot DF_n + DF_n)$$

$$= PMT_0(\sum DF) + DF_n - PMT_t(\sum DF) - DF_n = (PMT_0 - PMT_t)\sum DF$$

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A. Interest Rate Swaps/ 2 yrs. ago we entered into a swap for €100M, 7-yr. rec. fx., pay Libor with annual resets, swap rate was 2%

Maturity	DF
1	.99099
2	.977876
3	.965136
4	.951529
5	.937467
	4.822107

- at T_2: swap rate = 1.2968

rec. fx., pay. fl. T=0
- pay. fx., rec. fl. T=2

$$(r_{fx_0} - r_{fx_t})\sum DF \cdot NA$$

$\therefore V_{swap}$ = (.02 - .012968)4.822107 • 100M
= 3,390,905

$\begin{pmatrix} 3,375,000 \\ 1.3\% \end{pmatrix}$

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B. Currency Swaps/

£10M

\$14M @ 6%	Domestic Company (wants GBP)	6% in USD 5% in £	Foreign Company (wants USD)	£10M @ 5%

\$14M USD

- fixed-for-fixed

Pricing/
- we can view the swap as a pair of fixed rate bonds
- value of a currency swap = 0 when first negotiated

$V_{swap} = V_D - S_0V_F$

- the price of the swap is the set of fixed rates that result in $V_D = S_{d/f} \cdot V_F$

V_D – value of domestic bond
V_F – value of foreign bond
$S_0 = S_{d/f}$

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B. Currency Swaps/

e.g./ • US Company wants \$100M AUD for 1 yr.
• use 1 yr. currency swap with quarterly resets
• AUD/USD = 1.1400 (f/d) (30/360)

Rate	AUD	DF
L(90)	2.50	$1/(1+.025(90/360)) = .993789$
L(180)	2.60	$1/(1+.026(180/360)) = .987167$
L(270)	2.70	$1/(1+.027(270/360)) = .980152$
L(360)	2.80	$1/(1+.028(360/360)) = .972763$
		3.93387

$$r_{fx.AUD} = \frac{1-.972763}{3.93387} = .00692381$$
$$\times 4 = 2.7695\%$$

	USD	DF
L(90)	.10	$1/(1+.001(90/360)) = .999750$
L(180)	.15	$1/(1+.0015(180/360)) = .999251$
L(270)	.20	$1/(1+.002(270/360)) = .998502$
L(360)	.25	$1/(1+.0025(360/360)) = .997506$
		3.995009

$$r_{fx.USD} = \frac{1-.997506}{3.995009} = .00062422$$
$$\times 4 = .2497\%$$

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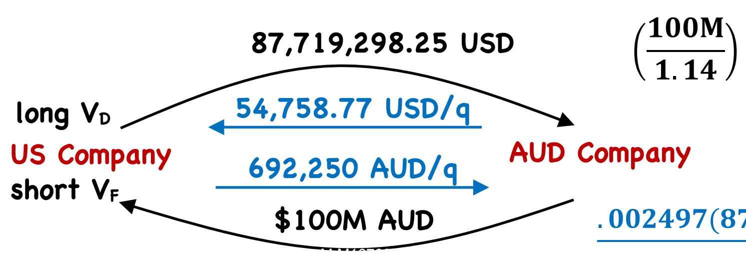
B. Currency Swaps/

e.g./ • US Company wants \$100M AUD for 1 yr.
• use 1 yr. currency swap with quarterly resets
• AUD/USD = 1.1400 (f/d) (30/360)

$\left(\frac{100M}{1.14}\right)$



$\frac{.02769(100M)}{4}$ $\frac{.002497(87,719,298.25)}{4}$

Valuation/

$V_{swap} = V_D - S_t V_F$ $= NA(r_{fx_d} \cdot \sum DF + DF_n) - S_t \cdot NA(r_{fx_f} \cdot \sum DF + DF_n)$

(in d) $S_t = d/f$

or $V_{swap} = S_t V_D - V_F$

(in f) $S_t = f/d$

V_{swap} = Payments to be received − Payments to be made

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B. Currency Swaps/

87,719,298.25 USD $\left(\frac{100M}{1.14}\right)$

54,758.77 USD/q ←

US Company — AUD Company

692,250 AUD/q →

$100M AUD

$\frac{.02769(100M)}{4}$ $\frac{.002497(87,719,298.25)}{4}$

60 days have passed (300 days left) $S_{AUD/USD} = 1.1300$

	AUD	DF
L(30)	2.00	.998336
L(120)	1.90	.993703
L(210)	1.80	.989609
L(300)	1.70	.986031
		3.967863

USD	DF
.50	.999584
.40	.998668
.30	.998253
.20	.998336
	3.994841

$$V_{swap} = V_D - S_t V_F = NA(r_{USD} \cdot \sum DF + DF_n)$$
(USD)

$$- S_{d/f} \cdot NA(r_{AUD} \cdot \sum DF + DF_n)$$
USD/AUD

$$= 87,719,298.25\left(\frac{.002497}{4} \cdot 3.994841 + .998336\right)$$

$$- {}^{1}/_{1.13} \cdot 100M\left(\frac{.02769}{4} \cdot 3.967863 + .986031\right)$$

$$= 87,792,085.92 - 89,690,135.54 = -1,898,049.62$$

$$= 99,205,057.09 - 101,349,853.20 = -2,144,049.62$$

$(-2,145,200)$

Page 30
LOS c, d
- describe
- compare
- calculate
- interpret

C. Equity Swap/

- OTC contract → one party pays a variable series determined by an equity
  → the other pays ① a variable rate series determined by another equity or rate
  ② a fixed series

a) receive equity, pay: floating / fixed / equity

b) pay equity, receive: floating / fixed / equity

⇒ Underlying for the equity leg → stock / index / custom portfolio

⇒ Equity leg cash flow can be with or without dividends

Page 31
LOS c, d
- describe
- compare
- calculate
- interpret

C. Equity Swap/

- Receive equity return, pay fixed   NA(equity r – fixed r)
  pay floating   NA(equity r – floating r)
  pay equity   NA(equity_a r – equity_b r)

e.g./ receive equity, pay fixed, quarterly resets (30/360)

NA = \$5M   (@ 1.6%/a)

- equity index return = 4% for Q_1

  NA(equity – fixed r) = 5M(.04 – .004) = 180,000

- equity index return = -6% for Q_2

  5M(-.06 – .004) = -320,000

⇒ Valuation/ $V_{swap} = V_{fx} - V_e$ - receive fixed, pay equity

or $V_{swap} = V_e - V_{fx}$ - receive equity, pay fixed

& fixed rate = $\dfrac{1 - DF_n}{\sum DF}$

Page 32
LOS c, d
- describe
- compare
- calculate
- interpret

C. Equity Swap/

receive fixed
pay equity
5 yr., annual resets (30/360)
r_{fx} = 1.5%
S_0 = 100
NA = \$10M

6 months later
r_{fx} = 1.2%   (all maturities = 1.2%
S_t = 105   - flat term structure)

Term	DF
.5	$1/\left(1+.012\left(180/360\right)\right) = .994036$
1.5	$1/\left(1+.012\left(540/360\right)\right) = .982318$
2.5	$1/\left(1+.012\left(900/360\right)\right) = .970874$
3.5	$1/\left(1+.012\left(1260/360\right)\right) = .959693$
4.5	$1/\left(1+.012\left(1620/360\right)\right) = .948767$
	4.855668

$$V_{swap} = V_{fx} - V_e$$
$$= NA(r_{fx} \cdot \sum DF + DF_n)$$
$$- NA\left(S_t/S_0\right)$$
$$= 10M(.015 \cdot 4.855668 + .948767)$$
$$- 10M\left(105/100\right)$$
$$= 10{,}216{,}020.20 - 10{,}500{,}000$$
$$= -283{,}979.80$$

Valuation of Contingent Claims

a. describe and interpret the binomial option valuation model and its component terms

b. calculate the no-arbitrage values of European and American options using a two-period binomial model

c. identify an arbitrage opportunity involving options and describe the related arbitrage

d. calculate and interpret the value of an interest rate option using a two-period binomial model

e. describe how the value of a European option can be analyzed as the present value of the option's expected payoff at expiration

f. identify assumptions of the Black-Scholes-Merton option valuation model

g. interpret the components of the Black-Scholes-Merton model as applied to call options in terms of a leveraged position in the underlying

h. describe how the Black-Scholes-Merton model is used to value European options on equities and currencies

i. describe how the Black model is used to value European options on futures

j. describe how the Black model is used to value European interest rate options and European swaptions

k. interpret each of the option Greeks

l. describe how a delta hedge is executed

m. describe the role of gamma risk in options trading

n. define implied volatility and explain how it is used in options trading

LOSs will match between the video and the PDFs, but may be in a different order than the CFAI readings

Valuation of Contingent Claims

Page 1/

Contingent claim - provides the owner a right (not an obligation) to a payoff determined by an underlying asset or rate

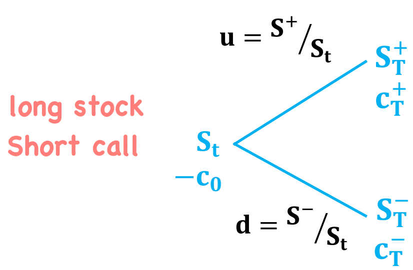
- options → valuation models (Binomial, BSM) based on the principle of no arbitrage
  - no price risk
  - no investment of personal funds

- valuation involves the following assumptions
  1/ replicating instruments are identifiable/investable
  2/ no market frictions
  3/ short selling allowed with full use of proceeds
  4/ underlying follows a known statistical distribution
  5/ borrow or lend at r_f

Note: LOS 1-5 will be done as one block since they don't separate well

Page 2/
LOS 1-5

Binomial Model (one and two periods)
↓
L1 Review



$\mathbf{u}$ and $\mathbf{d}$ are based on the volatility of the underlying

$$c = \max(0, S_T - X)$$

Step 1/ find a hedge ratio (h) to create a risk-free portfolio such that

$$hS^+ - c^+ = hS^- - c^-$$

$$hS^+ - hS^- = -c^- + c^+ \quad \rightarrow (c^+ - c^-)$$

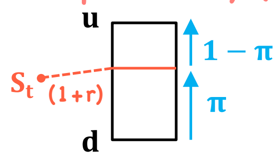
$$h(S^+ - S^-) = c^+ - c^-$$

$$h = \frac{c^+ - c^-}{S^+ - S^-}$$

Page 3/
LOS 1-5

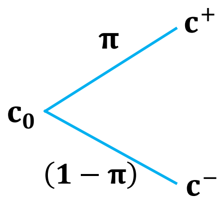
Step 2: determine a risk-neutral probability (π)

$$\pi = \frac{(1+r)-d}{u-d}$$



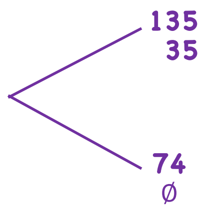
$\pi + (1-\pi) = 1$
or 100%

Step 3: Calculate c_0 (or p_0)



$$c_0 = \frac{\pi c^+ + (1-\pi)c^-}{(1+r)} \qquad p_0 = \frac{\pi p^+ + (1-\pi)p^-}{(1+r)}$$

e.g./ S_t = 100
x = 100
u = 1.35
d = .74
r_f = 5.15%



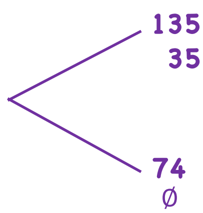
$$h = (35-0)/(135-74) = .57377$$

$$\pi = \frac{1.0515 - .74}{1.35 - .74} = .3115/.61 = .510656$$

$$c_0 = \frac{.510656(35)}{1.0515} = 16.998$$

Page 4/
LOS 1-5

e.g./ S_t = 100
x = 100
u = 1.35
d = .74
r_f = 5.15%



$h = .57377$
$\pi = .510656$
$c_0 = 16.998$

risk-free portfolio = long .57377 shares
short 1 call option

u: .57377(100)(1.35) = 77.45895
− 35
42.45895

d: .57377(100)(.74) = 42.45898

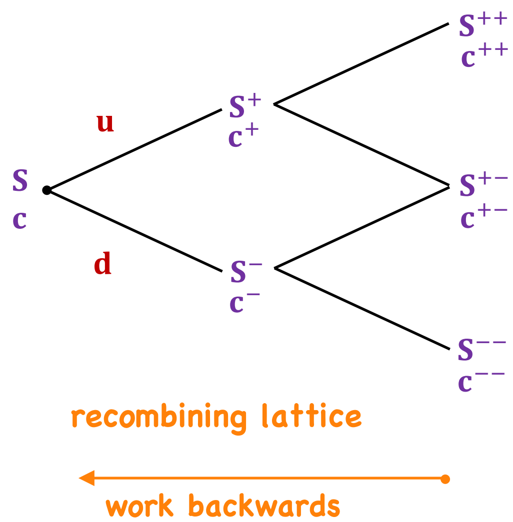
$$p_0 = \frac{(1-.510656)26}{1.0515} = 12.0998$$

or/ put-call parity

$$S_0 + p = c + PV(x)$$
$$p = c + PV(x) - S_0$$
$$= 16.998 + 100/1.0515 - 100$$
$$= 12.10$$

Binomial Option Valuation Model

Page 5/



$$\text{Max}(0, S^{++} - x) = \text{Max}(0, u^2 S - x)$$

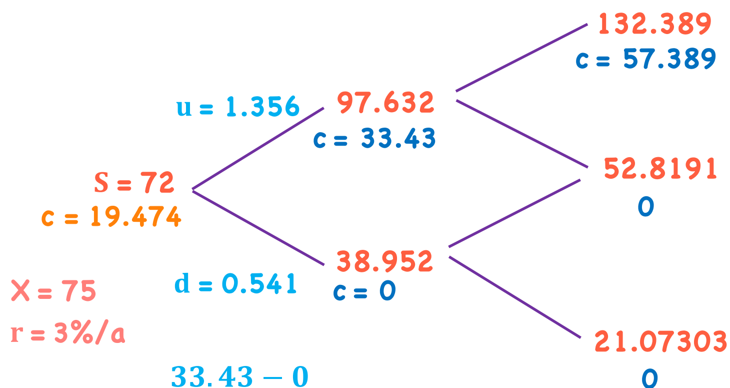
$$\text{Max}(0, S^{+-} - x) = \text{Max}(0, udS - x)$$

$$\text{Max}(0, S^{--} - x) = \text{Max}(0, d^2 S - x)$$

$$\Delta^+ = \frac{c^{++} - c^{+-}}{S^{++} - S^{+-}}$$

$$\Delta^- = \frac{c^{+-} - c^{--}}{S^{+-} - S^{--}}$$

Page 6/



$$\Delta = \frac{33.43 - 0}{97.632 - 38.952} = .5697$$

$$c = \frac{.6(33.43) + .4(0)}{1.03} = 19.474$$

$$\Delta^+ = \frac{57.389 - 0}{132.389 - 52.8191}$$

$$= .72124$$

$$\Delta^- = \frac{0 - 0}{S^{+-} - S^{--}} = 0$$

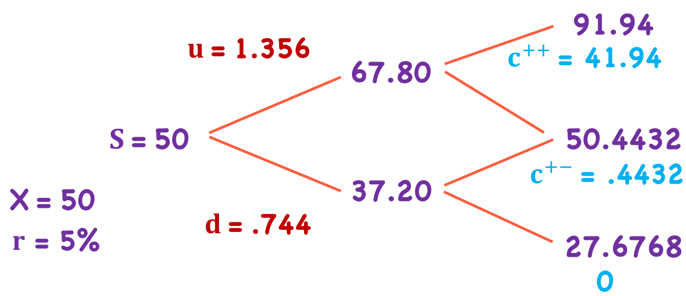
$$\pi = \frac{1.03 - .541}{1.356 - .541} = .60$$

$$c^+ = \frac{.6(57.389) + .4(0)}{1.03}$$

$$= 33.43$$

Page 7/

$$c = \frac{\pi^2 c^{++} + 2\pi(1-\pi)c^{+-} + (1-\pi)^2 c^{--}}{(1+r)^2}$$



$$\pi = \frac{1.05 - .744}{1.356 - .744} = .5$$

$$c = \frac{.5^2(41.94) + 2(.5)(.5).4432 + (.5)^2(0)}{(1.05)^2}$$

$$= \frac{10.485 + .2216}{1.05^2} = 9.7112$$

and/

$$S + p = \frac{X}{(1+r)^2} + c$$

$$50 + p = {}^{50}/_{(1.05)^2} + 9.7112$$

$$p = 45.351473 + 9.7112 - 50$$

$$= 5.0626$$

Page 8/

⇒ American Style Options/

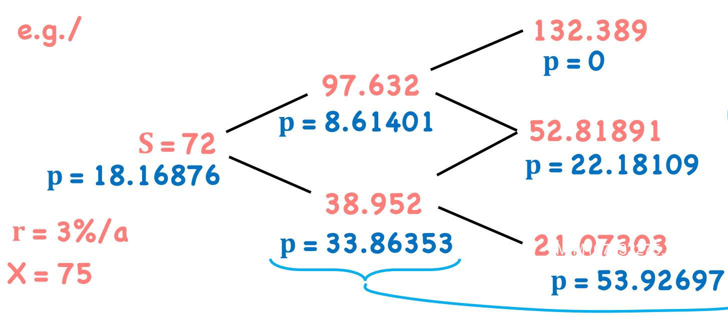
Calls ⇒ early exercise results in loss of time value

∴ better to sell than exercise

- no change in valuing c between European & American

Puts ⇒ early exercise results in a premium over European style options

e.g./



$$IV = X - S_t = 75 - 38.952$$

$$= 36.048$$

vs. 33.86353

- early exercise premium = 2.18447

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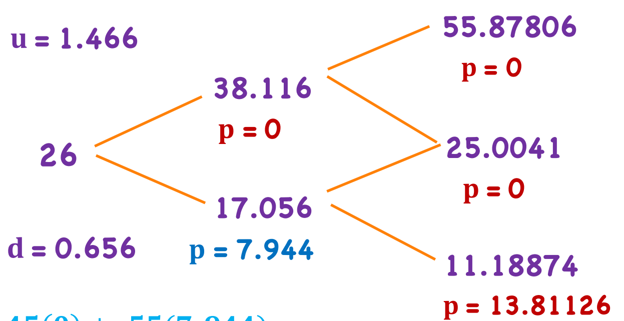
e.g./ $S_0 = 26$, $X = 25$, $u = 1.466$, $d = 0.656$, $r = 2.05\%$

- find exercise premium of put over European put price assuming a 2-period model



$$p = \frac{.45(0) + .55(7.944)}{1.0205} = 4.28143$$

premium = .26969

$$\pi = \frac{1.0205 - .656}{1.466 - .656} = 0.45$$

$$p = \frac{.45(0) + .55(13.81126)}{1.0205} = \cancel{7.4436}$$

$$p = \frac{.45(0) + .55(7.4436)}{1.0205} = 4.011739$$

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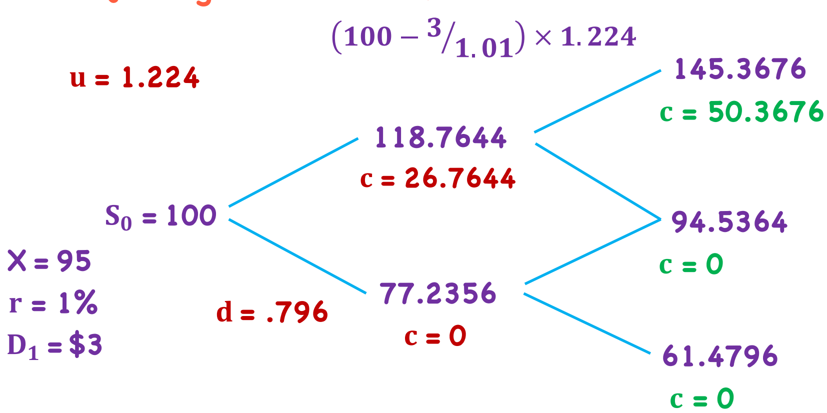
- Adjusting for dividends/



$$c^+ = \frac{.5(50.3676)}{1.01} = 24.934$$

Am. Style allows exercise before ex-div.

$$\pi = \frac{1.01 - .796}{1.224 - .796} = .50$$

$$c = \mathrm{Max}(0, S_T - X) = \mathrm{Max}(0, 118.7644 + 3 - 95) = 26.7644$$

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e.g./ S_0 = \$7.35   X = 8.00   put/calls ⇒ 2 yrs.

r = 4.35%   u = 1.445   d = 0.715

1) p & c @ T_0 & T_1 ⇒ European

$\underline{T_1}$

$$c^{++} = \text{Max}(0, u^2S - X) = \text{Max}(0, 1.445^2 \bullet 7.35 - 8) = 7.347$$

$$p^{++} = \text{Max}(0, X - u^2S) = \text{Max}(0, 8 - 1.445^2 \bullet 7.35) = 0$$

$$c^{+-} = \text{Max}(0, udS - X) = \text{Max}(0, 1.445(.715)(7.35) - 8) = 0$$

$$p^{+-} = \text{Max}(0, X - Sud) = \text{Max}(0, 8 - 7.35(1.445)(.715)) = .406$$

$$c^{--} = \text{Max}(0, d^2S - X) = \text{Max}(0, .715^2 \bullet 7.35 - 8) = 0$$

$$p^{--} = \text{Max}(0, X - Sd^2) = \text{Max}(0, 8 - 7.35(.715)^2) = 4.24$$

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e.g./ S_0 = \$7.35   X = 8.00   put/calls ⇒ 2 yrs.

r = 4.35%   u = 1.445   d = 0.715

1) p & c @ T_0 & T_1 ⇒ European

c^{++} = 7.347 p^{++} = 0

c^{+-} = 0 p^{+-} = .406

c^{--} = 0 p^{--} = 4.24

$$\pi = \frac{1.0435 - .715}{1.445 - .715} = 0.45$$

$$c^{+} = \frac{.45(7.347)}{1.0435} = 3.168$$

$$c^{-} = 0$$

$$p^{+} = \frac{.45(0) + .55(.406)}{1.0435} = .214$$

$$p^{-} = \frac{.45(.406) + .55(4.24)}{1.0435} = 2.41$$

$$c = \frac{(.45)^2(7.347) + 2(.45)(.55)(0) + (.55)^2 \bullet 0}{(1.0435)^2} = 1.488/(1.0435)^2 = 1.37$$

$$p = \frac{(.45)^2 \bullet 0 + 2(.45)(.55)(.406) + (.55)^2 4.24}{(1.0435)^2} = 1.484/(1.0435)^2 = 1.36$$

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e.g./ S_0 = \$7.35 X = 8.00 put/calls ⇒ 2 yrs.

r = 4.35% u = 1.445 d = 0.715

2) Δ @ T_1 & T_0 = European

c^{++} = 7.347 p^{++} = 0

c^{+-} = 0 p^{+-} = .406

c^{--} = 0 p^{--} = 4.24

$$S^{++} = 7.35(1.445)^2 = 15.347$$

$$S^{+-} = 7.35(1.445)(.715) = 7.594$$

$$S^{--} = 7.35(.715)^2 = 3.758$$

$$h_c^+ = \frac{c^{++} - c^{+-}}{S^{++} - S^{+-}} = \frac{7.347 - 0}{15.347 - 7.594}$$

$$= .9476$$

$$h_p^+ = \frac{p^{++} - p^{+-}}{S^{++} - S^{+-}} = \frac{0 - .406}{15.347 - 7.594}$$

$$= -.0523$$

$$h_c^- = \frac{c^{+-} - c^{--}}{S^{+-} - S^{--}} = \frac{0}{S^{++} - S^{--}} = 0$$

$$h_p^- = \frac{p^{+-} - p^{--}}{S^{+-} - S^{--}} = \frac{.406 - 4.24}{7.594 - 3.758} = -1$$

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e.g./ S_0 = \$7.35 , X = 8.00 put/call ⇒ 2 yrs.

r = 4.35% u = 1.445 d = 0.715

2) Δ @ T_1 & T_0 ⇒ European

$$h_c = \frac{c^+ - c^-}{S^+ - S^-}$$

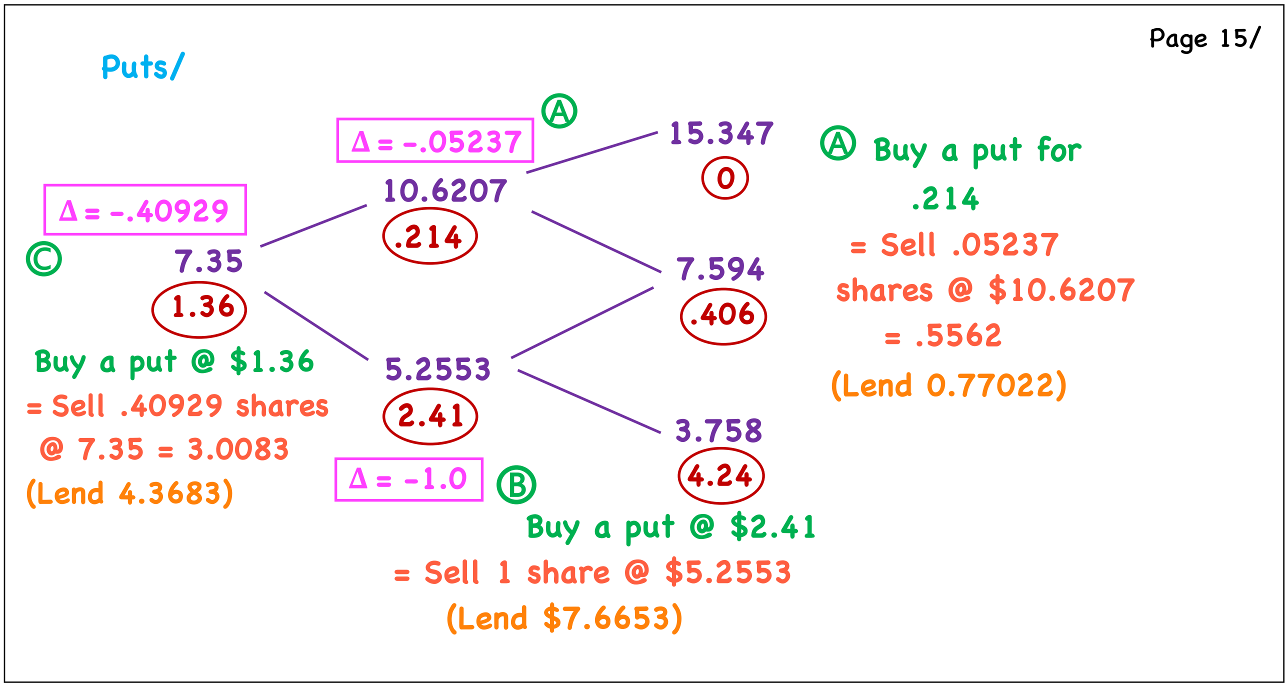
$$= \frac{3.16832 - 0}{10.6207 - 5.2553}$$

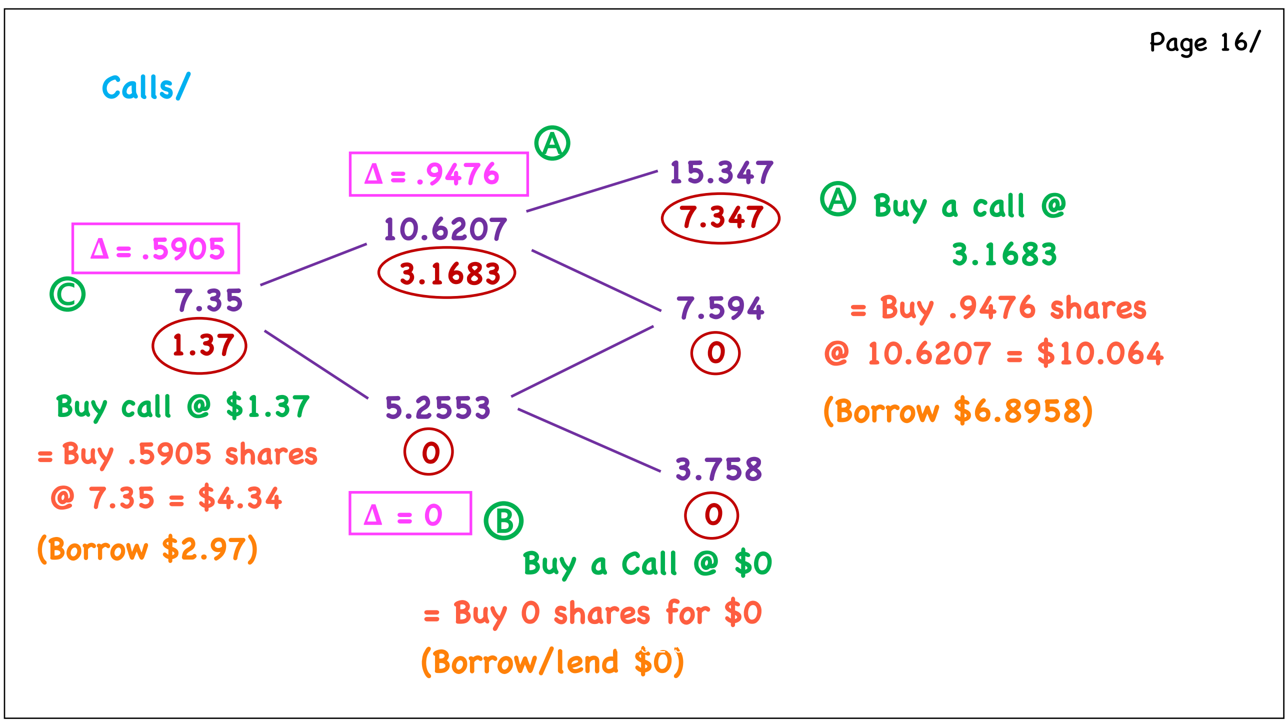
$$= .5905$$

$$h_p = \frac{p^+ - p^-}{S^+ - S^-}$$

$$= \frac{.2140 - 2.41}{10.6207 - 5.2553}$$

$$= -.40929$$

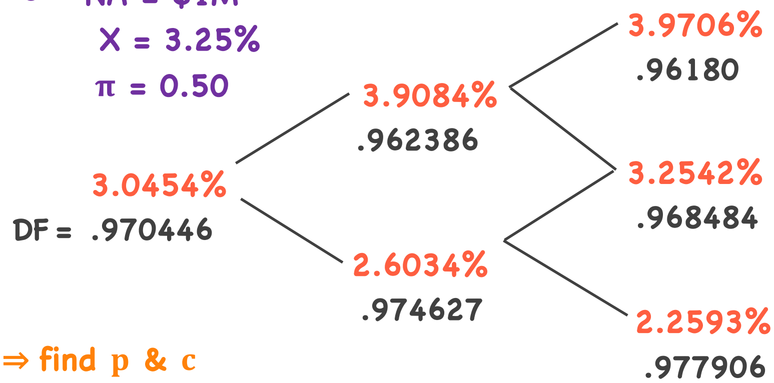

Page 15/
Puts/
Δ = -.05237
Ⓐ
15.347
0
Ⓐ Buy a put for
.214
= Sell .05237
shares @ $10.6207
= .5562
(Lend 0.77022)
Δ = -.40929
10.6207
.214
Ⓒ
7.35
1.36
7.594
.406
Buy a put @ $1.36
= Sell .40929 shares
@ 7.35 = 3.0083
(Lend 4.3683)
5.2553
2.41
3.758
4.24
Δ = -1.0
Ⓑ
Buy a put @ $2.41
= Sell 1 share @ $5.2553
(Lend $7.6653)


Page 16/
Calls/
Δ = .9476
Ⓐ
15.347
7.347
Ⓐ Buy a call @
3.1683
= Buy .9476 shares
@ 10.6207 = $10.064
(Borrow $6.8958)
Δ = .5905
10.6207
3.1683
Ⓒ
7.35
1.37
7.594
0
Buy call @ $1.37
= Buy .5905 shares
@ 7.35 = $4.34
(Borrow $2.97)
5.2553
0
3.758
0
Δ = 0
Ⓑ
Buy a Call @ $0
= Buy 0 shares for $0
(Borrow/lend $0)

Interest Rate Options

Page 17/

e.g./ NA = \$1M
X = 3.25%
π = 0.50



c^{++} = .007206
c^{+-} = .000042
c^{--} = 0
p^{++} = 0
p^{+-} = 0
p^{--} = .009907

⇒ find p & c

$$c^{+} = \frac{.5(.007206) + .5(.000042)}{1.039084}$$

$$= .003488$$

$$c^{-} = \frac{.5(.000042)}{1.026034}$$

$$p^{+} = {}^{0}/_{1.039084} = 0$$

$$p^{-} = \frac{.5(0) + .5(.009907)}{1.026034}$$

$$= .004828$$

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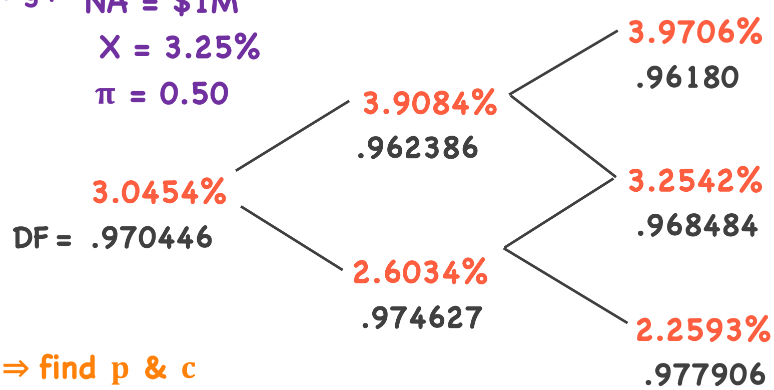
e.g./ NA = \$1M
X = 3.25%
π = 0.50



c^{++} = .007206
c^{+-} = .000042
c^{--} = 0
p^{++} = 0
p^{+-} = 0
p^{--} = .009907

⇒ find p & c

$$c = \frac{.5(.003488) + .5(.00002)}{1.030454} = .00170216$$

$$p = \frac{.5(0) + .5(.004828)}{1.030454} = .00234266$$

c^{+} = .003488
c^{-} = .00002
p^{+} = 0
p^{-} = .004828

BSM Assumptions

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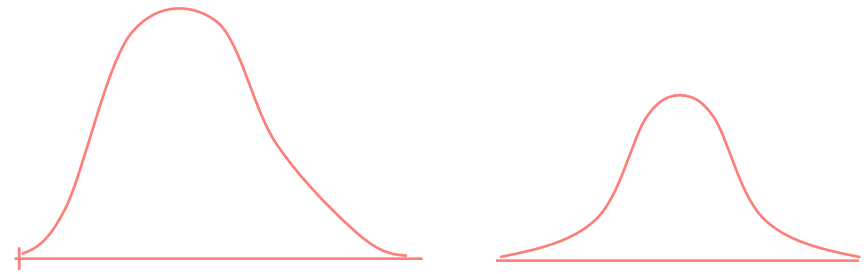
⇒ percentage changes in stock prices in a short period of time are approximately normally distributed

- changes in successive time periods are independent

∴ stock price itself has lognormal distribution



0 lognormal dist. normal dist.

bounded by zero

∴ $\underbrace{\ln S_T}$ is normal

↳ $\text{mean} = \ln S_0 + \left(\mu - \sigma^2/2\right)T$

↳ $\text{s.d.} = \sigma \cdot \sqrt{T}$

So,
$E(S_T) = S_0 e^{\mu T}$

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e.g./ S_0 = \$40 , $E(R)$ = 16%/annum
volatility = 20%/annum

- What is the prob. dist. of S_T in 6 mos.?

$$\ln S_T \sim N\left(\ln S_0 + \left[\mu - \sigma^2/2\right]T, \sigma\sqrt{T}\right)$$

$$\left(\ln 40 + \left[.16 - .2^2/2\right].5, (.2\sqrt{.5})\right)$$

$$\ln S_T \sim N(3.759, 0.141)$$

95% CI/

$$3.759 - (1.96 \times 0.141) < \ln S_T < 3.759 + (1.96 \times 0.141)$$

$$e^{3.759 - (1.96 \times 0.141)} < S_T < e^{3.759 + (1.96 \times .141)}$$

$$32.55 < S_T < 56.56$$

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Assumptions/

1. Stock prices are lognormally distributed
2. No transaction costs
3. No arbitrage opportunities
4. Trading is continuous
5. borrowing/lending at r_f , and r_f is constant
6. European options
7. Constant & known volatility of the underlying
8. Liquid underlying, divisible
9. Short selling with full use of proceeds
10. GBM describes price movements - continuous & smooth

BSM Model

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$$c = S \bullet N(d_1) - e^{-rT} X \bullet N(d_2)$$

$$(S - Xe^{-rT})$$

$$p = e^{-rT} X \bullet N(-d_2) - S \bullet N(-d_1)$$

$$(Xe^{-rT} - S)$$

- since the BSM is a continuous time model, r here is a continuously compounded rate

- $N(X)$ → represents risk-neutral probabilities

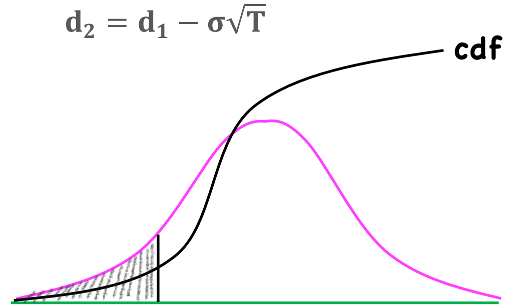
$r = r_f$

$N(\bullet)$

- standard normal cumulative distribution

$$d_1 = \frac{\ln\left(S/X\right) + \left(r + \sigma^2/2\right)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$



X = -1.645

then $N(X)$ = .05

$c = S \bullet N(d_1) - e^{-rT}X \bullet N(d_2)$ $\quad$ $p = e^{-rT}X \bullet N(-d_2) - S \bullet N(-d_1)$

⇒ BSM can be seen as having 2 components

Calls/ stock component $S \bullet N(d_1)$

bond component $e^{-rT}X \bullet N(d_2)$ or Xe^{-rT}

c = stock component − bond component

$S - B$

$S \bullet N(d_1)$ $\quad$ $Xe^{-rT} \bullet N(d_2)$

- goal is to replicate option payoffs with stocks & zero coupon bonds

Puts/ stock component $S \bullet N(-d_1)$

bond component $Xe^{-rT} \bullet N(-d_2)$

p = bond component − stock component

$B - S$

$Xe^{-rT} \bullet N(-d_2)$ $\quad$ $S \bullet N(-d_1)$

- cost of stock + bond portfolio = c or p

Calls/ cost $n_S \bullet S + n_B \bullet B$

$N(d_1) = n_S > 0$ $\quad$ $-N(d_2) = n_B < 0$

buy stock $\quad$ sell bonds

replicate payoff of a long call

- as S rises, $N(d_1)$ rises, as S drops, $N(d_1)$ drops

∴ must hedge by buying more stock in rising markets and sell stock in falling markets

- cumulative losses of buying high & selling low = c

∴ by 'Law of One Price', total cost of stock + bond portfolio = c

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e.g./ $S = 100$ $X = 100$ $r = 5\%$ $T = 1.0$ $\sigma = 30\%$

$$d_1 = \frac{\ln(S/X) + (r + \sigma^2/2)T}{\sigma\sqrt{T}} = \frac{\ln(1) + (.05 + .3^2/2)1.0}{.3\sqrt{1}}$$

$$d_2 = .3167 - .3 = .0167 \qquad = \frac{0 + .095}{.3} = .3167$$

$N(d_1) = N(.3167) = .624$ $[N \sim (0,1)]$

$N(d_2) = N(.0167) = .507$ - prob. that call option expires ITM

$N(-d_1) = N(-.3167) = 1 - .624 = .376$ $P(S_T > X)$

$N(-d_2) = N(-.0167) = 1 - .507 = .493$ - prob. that put option expires ITM

$$c = S \bullet N(d_1) - Xe^{-rT} \bullet N(d_2) = 100(.624) - 100e^{-.05}(.507) = 14.17$$

(buy shares) (sell bonds)

$$p = Xe^{-rT} \bullet N(-d_2) - S \bullet N(-d_1) = 100e^{-.05}(.493) - 100(.376) = 9.29$$

(buy bonds) (short shares)

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Replicate

	Buy a Call		Sell a Call	
Replicate	- buy $N(d_1)$ shares	S_T	- sell $N(d_1)$ shares	$-S_T$
	- sell $N(d_2)\%$ of a zero	$-X$	- buy $N(d_2)\%$ of a zero	$+X$
	$S \bullet N(d_1) - Xe^{-rT} \bullet N(d_2)$		$S \bullet - N(d_1) - Xe^{-rT} \bullet - N(d_2)$	
	Buy a Put		**Sell a Put**	
	- buy $N(-d_2)\%$ of a zero	X	- sell $N(-d_2)\%$ of a bond	$-X$
	- sell $N(-d_1)$ shares	$-S_T$	- buy $N(-d_1)$ shares	$+S_T$
	$Xe^{-rT} \bullet N(-d_2) - S \bullet N(-d_1)$		$Xe^{-rT} \bullet - N(-d_2) - S \bullet - N(-d_1)$	

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⇒ modifications for carry benefits/

$c = Se^{-\gamma T} \cdot N(d_1) - Xe^{-rT} \cdot N(d_2)$

$p = Xe^{-rT} \cdot N(-d_2) - Se^{-\gamma T} \cdot N(-d_1)$

$$d_1 = \frac{\ln(S/X) + \left(r - \gamma + \sigma^2/2\right)T}{\sigma\sqrt{T}} \qquad d_2 = d_1 - \sigma\sqrt{T}$$

lowers d_1

∴ lowers d_2 - the prob. $(S_T - X > 0)$

- the prob. of the call being ITM drops as γ rises

⇒ using put-call parity/

$$Se^{-\gamma T} + p = Xe^{-rT} + c$$

γ - carry benefits expressed as a continuous yield

Note: carry benefits lower the value of the underlying

- reduces value of calls
- increases value of puts

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⇒ modifications for carry benefits/

- BSM applied to equities $(\gamma = \delta)$

  $c = Se^{-\delta T} \cdot N(d_1) - Xe^{-rT} \cdot N(d_2)$

  $p = Xe^{-rT} \cdot N(-d_2) - Se^{-\delta T} \cdot N(-d_1)$

- BSM applied to currencies $(\gamma = r^f)$

  $c = Se^{-r_f T} \cdot N(d_1) - Xe^{-rT} \cdot N(d_2)$

  $p = Xe^{-rT} \cdot N(-d_2) - Se^{-r_f T} \cdot N(-d_1)$

Black Model

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$c = [F_0(T) \cdot N(d_1) - X \cdot N(d_2)]e^{-rT}$

$F_T \quad - \quad X$

$p = [X \cdot N(-d_2) - F_0 \cdot N(-d_1)]e^{-rT}$

$X \quad - \quad F_0$

$$d_1 = \frac{\ln\left(\frac{F_0(T)}{X}\right) + \left(\frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

CL - NOV. 17 - \$50 (1000)

call → \$55 cheap/ (X)

e.g./	Calls	Puts
	$N(d_1) = .491$	$N(-d_1) = .509$
	$N(d_2) = .461$	$N(-d_2) = .539$
	$c = \$51.41$	$p = \$59.76$

Call = $PV(.491 \times F_0(T) - .461 \times X)$

Put = $PV(.539 \times X - .509 \times F_0(T))$

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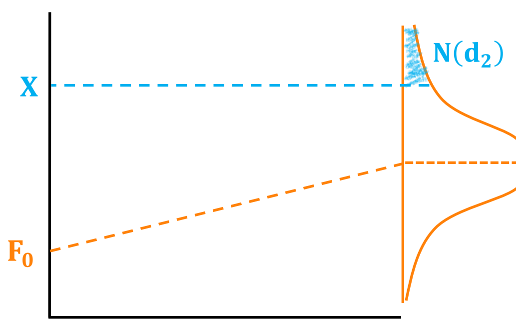
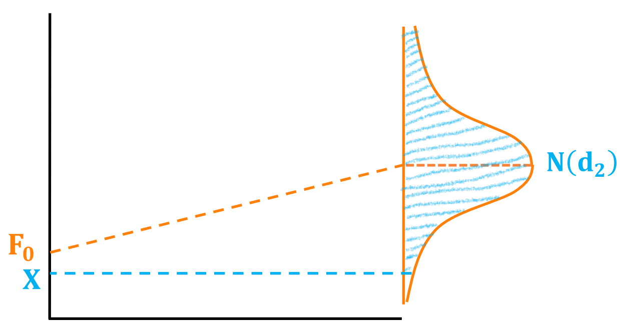


- the further out of the money the strike, the lower the probability of exercise

  (i.e. ⇒ lower $N(d_2)$)



- in the money call, higher probability of exercise

Black Model - Interest Rates

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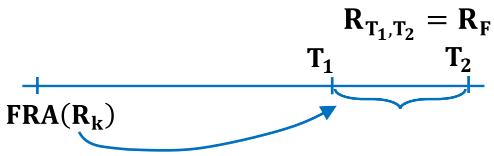

- interest rate call option

⇒ underlying = FRA

⇒ underlying = Libor rate

$$c = AP \cdot [\underbrace{FRA}_{FRA_{T_1,T_2}} \cdot N(d_1) - \underbrace{X_R}_{\text{exercise rate}} \cdot N(d_2)]\underbrace{e^{-rT}}_{\text{accrual period of option}}$$

$T \rightarrow T_2$

$$p = AP \cdot [X_R \cdot N(-d_2) - FRA \cdot N(-d_1)]e^{-rT}$$

$$d_1 = \frac{\ln\left(FRA_{T_1,T_2}/X_R\right) + \left(\sigma^2/2\right)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

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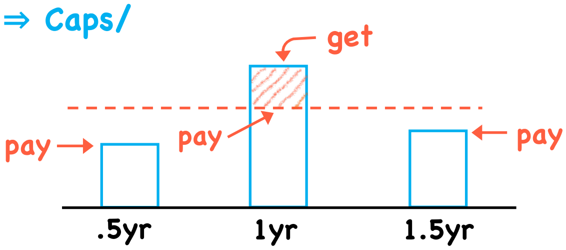
⇒ if X_R = FRA_0

long FRA = borrower

long call + short put } receive floating, pay fixed

long put + short call } receive fixed, pay floating

⇒ Caps/



caplet + caplet + caplet = Cap

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⇒ Caps/

e.g./

(up - up)

NA = $100
X = 5.75%
(semi)

Time 0 1 2

At T = 0, r = 5.54% , X_R ⇒ OTM — Company receives $0, pays $2.77

At T = 1, r = 6.004%, X_R ⇒ ITM — Company receives 0.127, pays 3.002, net $ 2.875

At T = 2, r = 6.915%, X_R ⇒ ITM — Company receives .5825, pays 3.4575, net $ 2.875

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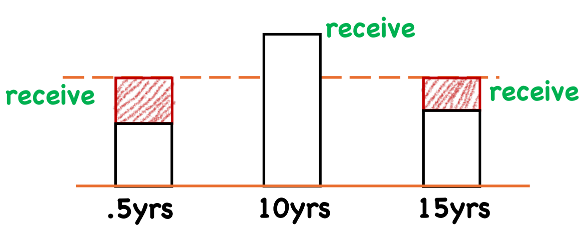
⇒ Floors/ puts



floorlet + floorlet + floorlet = Floor

- a floating rate investment with a floor

⇒ long a floor – receive fixed
short a cap – pay floating
} fixed for floating swap

⇒ if X_R = swap rate , $V_{cap} = V_{floor}$ ∴ cost = $0

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⇒ Swaptions/ - option on a swap
- holder ⇒ right to enter a swap at a pre-agreed swap rate

Payer swaption - pay fx., get fl.

fixed

Receiver swaption - get fx., pay fl.

(fixed)
- if swap rate increases before expiration, buyer enters into a pay fx. @ X_R get fl.
⇒ can enter an offsetting receive fixed-pay floating swap
- floating legs offset
- annuity = fixed rate - X_R

$$PV_A = \sum_{j=1}^{n} (r_{fix} - X_R) \cdot NA$$

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$$Pay_{swn} = [R_{fix} \cdot N(d_1) - X_R \cdot N(d_2)] \cdot PV_A$$

$$Rec_{swn} = [X_R \cdot N(-d_2) - R_{fix} \cdot N(-d_1)] \cdot PV_A$$

$$d_1 = \frac{\ln\left(R_{fix}/X_R\right) + \left(\sigma^2/2\right)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

When swaption expires

e^{-rT}? where is the discount factor?
- since the payment is a series of payments, the PV_A embeds the discount factor

Option Greeks

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Delta/ - change in the value of the option for a change in the price of the underlying

underlying itself has delta = 1

$\text{Delta}_c = e^{-\delta T} \cdot N(d_1)$

$\text{Delta}_p = e^{-\delta T} \cdot N(-d_1)$

- for a given X, $\text{delta}_c + \text{delta}_p = 1.0$

∴ long call, short put at some X

$= e^{-\delta T} \cdot N(d_1) + -e^{-\delta T} \cdot N(-d_1)$

$= e^{-(\delta T - \delta T)} \left(N(d_1) + N(-d_1)\right)$

$= e^{0}\ (1)$

$= 1 \times 1 = 1$

→ a $T \to 0$, delta → 0 or 1 (OTM vs. ITM)

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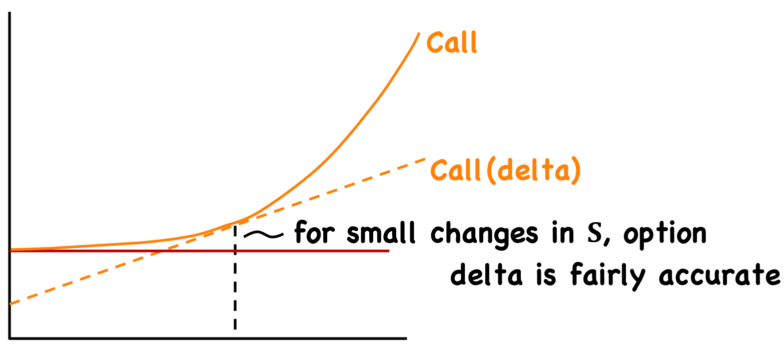
Delta/



- for larger changes, delta loses accuracy

- delta hedging ⇒ taking an offsetting delta position so as to be delta neutral

e.g./ long 100 shares = long 100 deltas
sell 2 ATM calls = short (50 x 2) deltas

protects from small moves up/down in S.

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- delta hedging/
  - delta neutral implies

(portfolio delta + N_H $Delta_H$) = 0

e.g. 100 + -2 (50) = 0

$$N_H = \frac{-\text{portfolio delta}}{Delta_H} = {}^{-100}/_{50} = -2$$

e.g./ $S = 100$ $X = 100$ $r = 5\%$ $T = 1.0$ $\sigma = 30\%$ $\delta = 5\%$

position: short puts on 10,000 shares

$Delta_c$ = 0.532 delta hedge = short 4,190 shares

$Delta_p$ = -0.419 or

delta hedge = ${}^{-4190}/_{.532} = -7{,}876$ calls

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Gamma/ - the change in an option's delta for a change in S_0

e.g./ $S_0 = 50$, $X = 50$ delta = .50 gamma = .03

as $S_0 \rightarrow 51$, $c\uparrow$ .50 , but as $S_0 \rightarrow 52$, $c\uparrow$ .53

- a measure of the curvature in option price vs. the stock price

$$gamma_c = gamma_p = \frac{e^{-\delta T}}{S \bullet \sigma\sqrt{T}}\ n(d_1) \Rightarrow \text{pdf, not cdf.}$$

i.e. at S & X , $delta_c + delta_p$ = 1.0

at $S+G$ & X , $delta_{c+g} + delta_{p-g}$ = 1.0

let g = change in delta

G = change in S

- gamma of underlying = 0, since delta is constant at 1 (the underlying will always move 1-for-1 with itself)

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Gamma/ - gamma is always non-negative
& is largest ATM
- gamma very small when deep ITM/OTM

- gamma measures the risk that remains once a portfolio is delta neutral
- gamma risk can be managed to a specific level but never eliminated

e.g./ long 100 shares = 100 deltas

a) sell 2 $\underbrace{\text{ATM calls}}_{\text{highest gamma}}$ = short 100 deltas

b) sell $\underbrace{\text{deep ITM calls}}_{\text{lowest gamma}}$/ e.g. - delta_c = .80

- Sell 2 calls
  Buy 60 shares
- Sell 1 call
  Sell 20 shares

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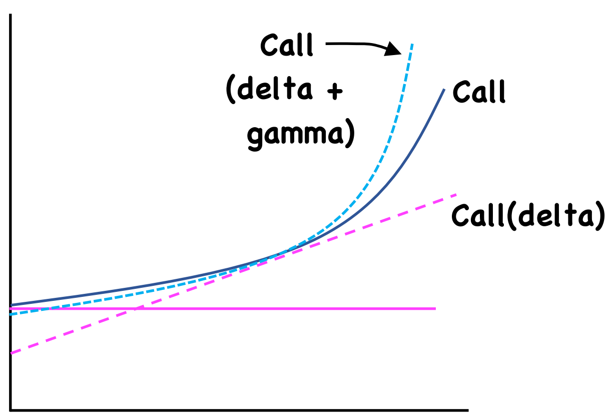
Gamma/

$$c_{new} - c_{old} \approx (S_{new} - S_{old})\text{delta}_c + \frac{\text{gamma}_c}{2}(S_{new} - S_{old})^2$$

$$p_{new} - p_{old} \approx (S_{new} - S_{old})\text{delta}_p + \frac{\text{gamma}_p}{2}(S_{new} - S_{old})^2$$



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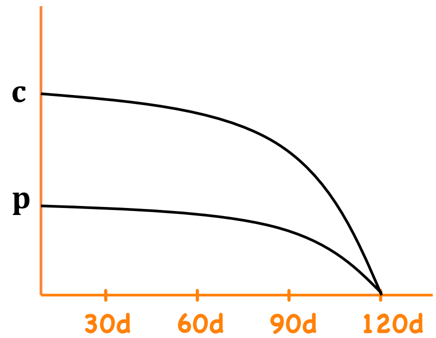
⇒ Theta/ - a change in c & p for a change in time to expiration

long options = negative theta (time is your enemy)

short options = positive theta (time is your friend)

- theta = rate of option decay



- theta cannot be adjusted by positions in the underlying

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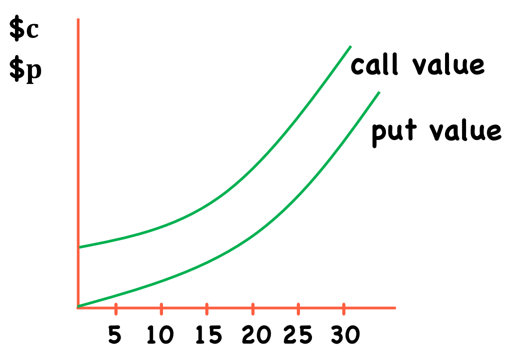
⇒ Vega/ - change in c & p for a change in volatility

long options = positive vega (long volatility)

short options = negative vega (short volatility)

- vega is based on future volatility (unobservable)

- option value is very sensitive to vega



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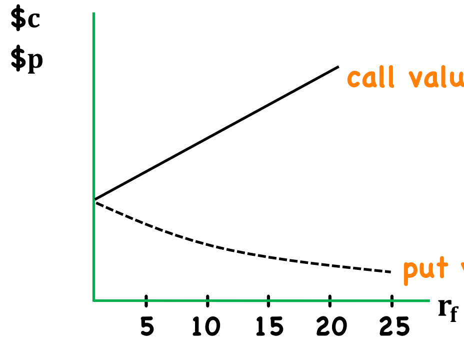
⇒ Rho/ – a change in c & p for a change in the risk-free rate

calls – positive rho (S_0e^{rT}) – raises cost of carry as r_f ↑

puts – negative rho (Xe^{-rT}) – lowers PV as r_f ↑



Implied Volatility

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⇒ volatility can be estimated from historical data

- in BSM, vol. is 'future volatility'

- volatility can be inferred from prices ⇒ implied volatility
  - common to see different implied vols. for different
    - → exercise prices
      - volatility smile/skew
    - → times to expiration
      - term structure of volatility
    - → positions with the same terms (puts & calls)
- options are sometimes quoted in volatilities

	Belief
e.g./ 3-mos. ATM call @ 19% IV on an index	25%
1-mos. ITM put @ 24% IV on a component	20%

Action: Buy index Call, sell component put
(long vol.) (short vol.)

REVIEW

Pricing & Valuation of Forward Commitments

Review – 1

- general:

1) no underlying cash flows: $F_0 = S_0(1 + r)^T$ $\quad F_0 = S_0e^{rT}$

$V_t = \dfrac{F_t - F_0}{(1 + r)^{T-t}}$ $\quad V_t = (F_t - F_0)e^{-r(T-t)}$

2) underlying cash flows: $F_0 = [S_0 + PV(\theta) - PV(\gamma)](1 + r)^T$

$\$$ $\quad \$$

3) underlying with known yield $F_0 = S_0e^{(r + \theta - \gamma)T}$

%

A/ Equities → all equations above

B/ Interest Rates → Forward Rate Agreement

6x9 FRA

0 — 6 — 9

6: expires

9: payoff

discounted back

- advanced set
advanced settled

Review – 2

B/ Interest Rates → Forward Rate Agreement

- rec. fl.

$$\frac{NA(\text{fl. rate} - \text{fx. rate})\left({}^{days}/_{360}\right)}{\left[1 + \text{fl. rate}\left({}^{days}/_{360}\right)\right]}$$

- rec. fx.

$$\frac{NA(\text{fx. rate} - \text{fl. rate})\left({}^{days}/_{360}\right)}{\left[1 + \text{fl. rate}\left({}^{days}/_{360}\right)\right]}$$

C/ Fixed Income

Quoted Futures Price (QF_0) $\quad QF_0 = {}^{F_0}/_{CF}$

↓

- generic bond

deliverable bonds

CTD

↓

$$F_0 = \left[(S_0 + AI) - {}^{Int.}/_{(1 + r)^t}\right](1 + r)^T - AI$$

$F_0 = QF_0 \times CF$

Review - 3

D. Currency Forwards r_f - foreign int. rate
r - domestic int. rate

$$F_{f/d} = S_{f/d}\frac{(1+r_f)^T}{(1+r)^T} \quad \text{or} \quad F_{d/f} = S_{d/f}\frac{(1+r)^T}{(1+r_f)^T}$$

$$\text{or} \quad F_{f/d} = S_{f/d}e^{(r_f-r)T} \qquad F_{d/f} = S_{d/f}e^{(r-r_f)T}$$

$$V_t = \frac{(F_0 - F_T)NA}{(1+r)^{T-t}} \text{ (short)} \qquad V_t = \frac{(F_t - F_0)NA}{(1+r)^{T-t}} \text{ (long)}$$

1. Interest Rate Swaps fixed rate = $\frac{1 - DF_n}{\sum DF}$

$$V_{swap} = V_{fx_0} - V_{fx_t}$$
$$= NA(r_{fx_0} - r_{fx_t})\sum DF$$

Review - 4

2. Currency Swaps - 2 fixed rates

$$r_{dom} = \frac{1 - DF_n}{\sum DF} \qquad r_{for} = \frac{1 - DF_n}{\sum DF}$$

$V_{swap} = V_D - S_{d/f}V_F$ - in domestic currency

$$= NA(r_{fx_d} \cdot \sum DF + DF_n) - S_t \cdot NA(r_{fx_f} \cdot \sum DF + DF_n)$$

$V_{swap} = S_{f/d} \cdot V_D - V_F$ - in foreign currency

$$= S_t \cdot NA(r_{fx_d} \cdot \sum DF + DF_n) - NA(r_{fx_f} \cdot \sum DF + DF_n)$$

3. Equity Swaps NA (equity r - fixed r) - rec. equity, pay fx.

$$V_{swap} = V_e - V_f = NA\left({S_t}/{S_0}\right) - NA(r_{fx} \cdot \sum DF + DF_n)$$

or

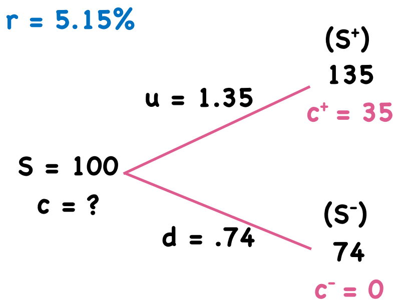
$$V_{swap} = V_f - V_e = NA(r_{fx} \cdot \sum DF + DF_n) - NA\left({S_t}/{S_0}\right)$$

(rec. fx., pay equity)

Valuation of Contingent Claims

Review - 1

⇒ Binomial Option Valuation Model/ no-arbitrage



$$\Delta = \frac{c^+ - c^-}{S^+ - S^-} = \frac{35 - 0}{135 - 74} = .5737705$$

$$p = \Delta S + PV(-\Delta S^+ + p^+)$$

$$c = \Delta S + PV(-\Delta S^+ + c^+)$$

$$= .5737705(100) + \frac{(-.5737705(135) + 35)}{1.0515}$$

$$= 16.9975$$

= Alternative/

$$\pi = \frac{(1+r) - d}{u - d} = \frac{1.0515 - .74}{1.35 - .74} = .510656$$

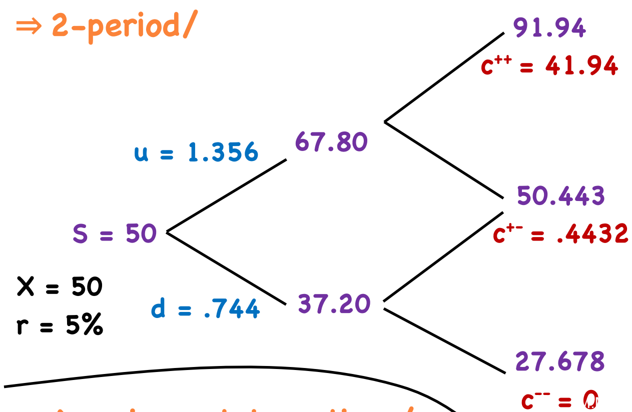
$$c = PV[\pi c^+ + (1-\pi)c^-] = \frac{.510656(35) + (1 - .510656)0}{1.0515} = 16.9975$$

$$p = PV[\pi p^+ + (1-\pi)p^-] = \frac{.510656(0) + (1 - .510656)(26)}{1.0515} = 12.10$$

⇒ Put-Call Parity/ $S + p = \frac{x}{(1+r)} + c$ / $100 + p = \frac{100}{1.0515} + 16.9975$ / $p = 12.10$

Review - 2

⇒ 2-period/



$$\pi = \frac{1.05 - .744}{1.356 - .744} = .50$$

$$c = \frac{\pi^2 c^{++} + 2\pi(1-\pi)c^{+-} + (1-\pi)^2 c^{--}}{(1+r)^2}$$

$$= \frac{.5^2(41.94) + 2(.5)(.5)(.4432) + (.5)^2(0)}{(1.05)^2}$$

$$= 9.7112$$

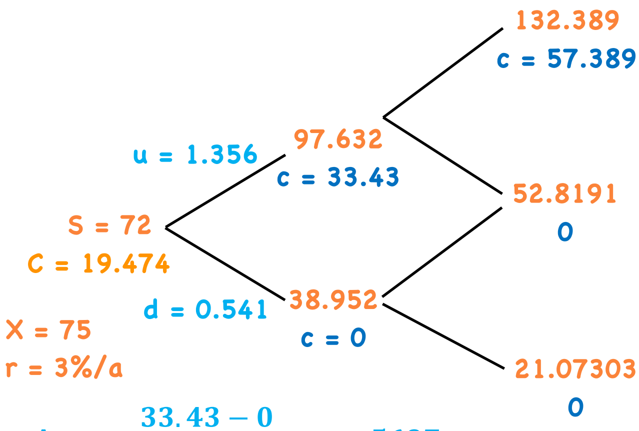
$$S + p = \frac{X}{(1+r)^2} + c$$

$$50 + p = \frac{50}{(1.05)^2} + 9.7112 = 5.0626$$

⇒ American style options/

- call ⇒ no change
- always better to sell than to exercise early
- put ⇒ early exercise results in a premium over European puts.

Binomial Option Valuation Model



$$\Delta = \frac{33.43 - 0}{97.632 - 38.952} = .5697$$

$$c = \frac{.6(33.43) + .4(0)}{1.03} = 19.474$$

$$\Delta^{+} = \frac{c^{++} 57.389 - 0}{132.389 - 52.8191}$$

$$= .72124$$

$$\Delta^{-} = \frac{0 - 0}{S^{+-} - S^{--}} = 0$$

$$\pi = \frac{1.03 - .541}{1.356 - .541} = .60$$

$$c^{+} = \frac{.6(57.389) + .4(0)}{1.03}$$

$$= 33.43$$

Valuation of Contingent Claims

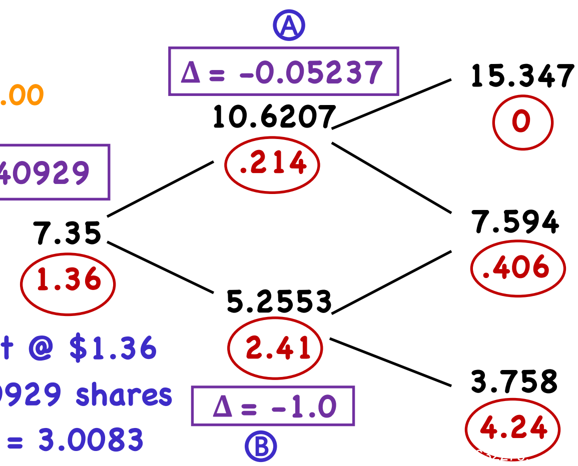
Review - 3

Puts/

x = 8.00

Δ = -.40929

Ⓒ

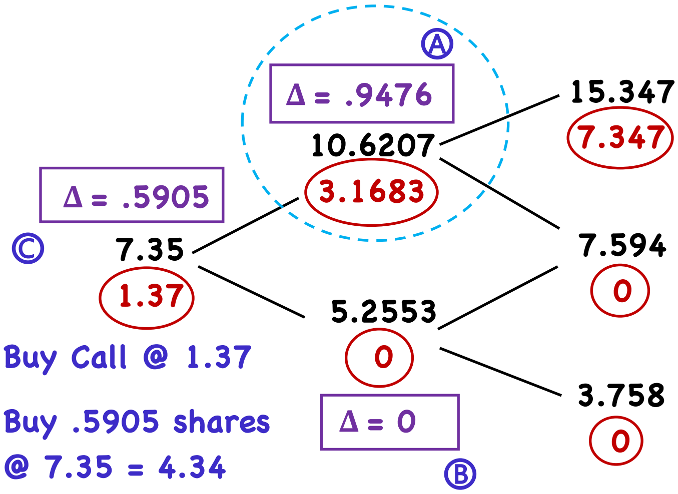


Buy put @ $1.36
Sell .40929 shares
@ 7.35 = 3.0083
(Lend 4.3683)

Buy put @ $2.41
Sell 1 shares @ $5.2553
(Lend 7.6653)

Ⓐ Buy put for
0.214
Sell .05237
shares @ $10.6207
= .5562
(Lend 0.77022)

Review - 4



Buy Call @ 1.37

Buy .5905 shares
@ 7.35 = 4.34
(Borrow $2.97)

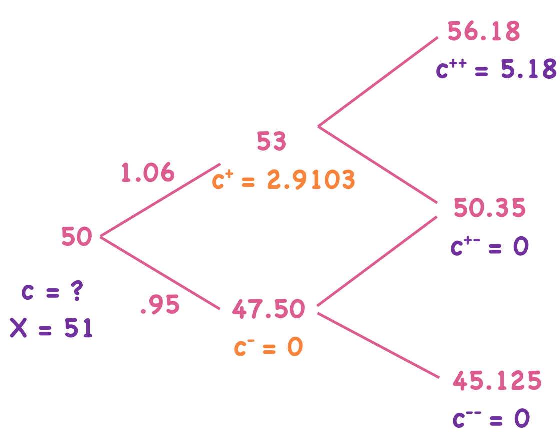
Ⓐ Buy Call @
$3.1683

Buy .9476 shares
@ 10.6207 = $10.064
(Borrow $6.8958)

Buy Call @ 0
Buy 0 shares for $0
Borrow/lend $0

1. A stock price is currently $50. Over each of the next two three-month periods it is expected to go up by 6% or down by 5%. The risk-free interest rate is 5% per annum with continuous compounding. What is the value of a six-month European call option with a strike price of $51.

$$\frac{(1+r)-d}{u-d}$$



$$\pi = \frac{e^{.05(.25)} - .95}{1.06 - .95} = .5689$$

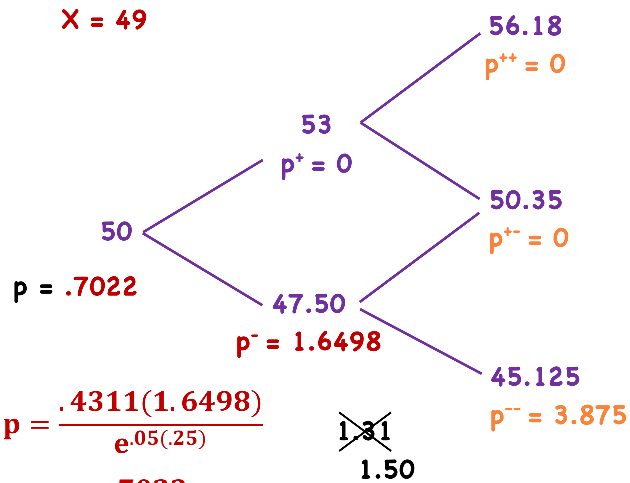
$$c^{+} = \frac{.5689(5.18)}{e^{.05(.25)}} = 2.9103$$

$$(\pi c^{+} + (1-\pi)c^{-})$$

$$c = \frac{.5689(2.9103)}{e^{.05(.25)}} = 1.6351$$

2. A stock price is currently \$50. Over each of the next two three-month periods it is expected to go up by 6% or down by 5%. The risk-free interest rate is 5% per annum with continuous compounding. What is the value of a six-month American put option with a strike price of \$49.

X = 49



$$p = \frac{.4311(1.6498)}{e^{.05(.25)}}$$
$$= .7022$$

$$\pi = \frac{e^{.05(.25)} - .95}{1.06 - .95} = .5689$$

$$p^{+} = 0$$

$$p^{-} = \frac{.5689(0) + .4311(3.875)}{e^{.05(.25)}}$$
$$= 1.6498$$

Review - 7

⇒ BSM/ - from discrete time (binomial) to continuous time

$$c = S\,N(d_1) - e^{-rT}X\,N(d_2)$$
$$S - Xe^{-rT}$$

$N(\cdot)$ - standard normal cumulative distribution

$$p = e^{-rT}X\,N(-d_2) - S\,N(-d_1)$$
$$Xe^{-rT} - S$$

$$d_1 = \frac{\ln\left(S/X\right) + \left(r + \sigma^2/2\right)T}{\sigma\sqrt{T}}$$

risk-free rate

$$d_2 = d_1 - \sigma\sqrt{T}$$

$\{d_1, d_2\}$ risk neutral probabilities

e.g./ S = 100, X = 100, r = 5% T = 1.0 σ = 30%

$$d_1 = \frac{\ln(1) + \left(.05 + .3^2/2\right)1.0}{.3\sqrt{1}} = .3167$$

$$d_2 = .3167 - .3\sqrt{1} = 0.0167$$

$N(d_1) = N(.3167) = .624$

$N(d_2) = N(.0167) = .507$ - prob. that c expires ITM

$N(-d_1) = N(-.3167) = 1 - .624 = .376$

$N(-d_2) = N(-.0167) = 1 - .507 = .493$

$$c = 100(.624) - 100e^{-.05}(.507) = 14.17$$

buy shares sell bonds

$$p = 100e^{.05}(.493) - 100(.376)$$

buy bonds sell shares

$$= 9.29$$

Review - 8

	Buy a Call		Sell a Call	
Replicate	- buy $N(d_1)$ Shares	S_T	- sell $N(d_1)$ shares	$-S_T$
	- sell $N(d_2)$% of a zero	$-X$	- buy $N(d_2)$% of a zero	$+X$
c =	$S \cdot N(d_1) - Xe^{-rT} N(d_2)$		$S - N(d_1) - Xe^{-rT} - N(d_2)$	
	Buy a Put		**Sell a Put**	
	- buy $N(-d_2)$% of a bond	X	- sell $N(-d_2)$% of a bond	$-X$
	- sell $N(-d_1)$ shares	$-S_T$	- buy $N(-d_1)$ shares	S_T
p =	$Xe^{-rT} N(-d_2) - S\,N(-d_1)$		$Xe^{-rT} - N(-d_2) - S \cdot -N(-d_1)$	

⇒ modification for carry benefits/ γ - carry benefits (continuous comp.)

(dividends $\gamma = \delta$)

(currencies $\gamma = r_f$)

$$c = Se^{-\gamma T} N(d_1) - Xe^{-rT}(d_2) \qquad d_1 = \frac{\ln(S/X) + \left(r - \gamma + \sigma^2/2\right)T}{\sigma\sqrt{T}}$$

Review - 9

⇒ Black Model/

F_0 ⇒ futures contract

$$c = [F_0(T) \cdot N(d_1) - X \cdot N(d_2)]e^{-rT}$$

$$p = [X \cdot N(-d_2) - F_0 \cdot N(-d_1)]e^{-rT}$$

- underlying is costless to carry

$$d_1 = \frac{\ln\left(\frac{F_0}{X}\right) + \left(\frac{\sigma^2}{2}\right)T}{\sigma\sqrt{2}} \qquad d_2 = d_1 - \sigma\sqrt{T}$$

FRA - forward rate agreement

$$c = AP[\underbrace{FRA}_{FRA_{T_1,T_2}} \cdot N(d_1) - \underbrace{X_R}_{\text{exercise rate of the call}} \cdot N(d_2)]e^{-rT} \sim T_2$$

$$p = AP[X_R\, N(-d_2) - FRA \cdot N(-d_1)]e^{-rT}$$

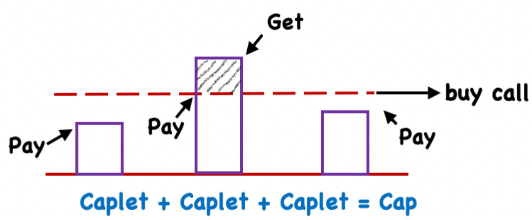
AP - accrual period scaling variable

$$d_1 = \frac{\ln\left(\frac{FRA_{T_1,T_2}}{X_R}\right) + \left(\frac{\sigma^2}{2}\right)T_1}{\sigma\sqrt{T_1}} \qquad d_2 = d_1 - \sigma\sqrt{T_1}$$

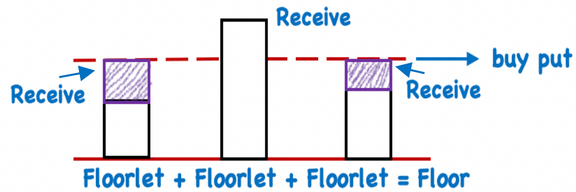
Review - 10

⇒ Black Model/

FRA/ Caps



Floors



long a floor + short a cap } fixed for floating

& if X_R = swap rate, $V_{cap} = V_{floor}$

∴ cost = 0

⇒ Swaptions/ - option on a swap

payer swaption - pay fx., get fl.

receiver swaption - pay fl., get fx.

Review - 11

⇒ Option Greeks/

delta - change in the value of the option for a change in the price of the underlying

- for a given X, $|\text{delta}_c| + |\text{delta}_p| = 1.0$
- on expiration date, delta = 1 (ITM) or 0 (OTM)

delta hedging/ S = 100 X = 100

$\text{Delta}_c = .532$ $\text{Delta}_p = -.419$

position: short puts on 10,000 shares

① short 4,190 shares

② short $\frac{-4190}{.532} = -7,876$ calls

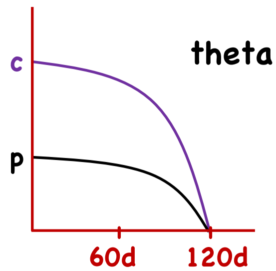
gamma/ change in delta for a change in underlying

$\text{gamma}_c = \text{gamma}_p$

Review - 12

⇒ Option Greeks/

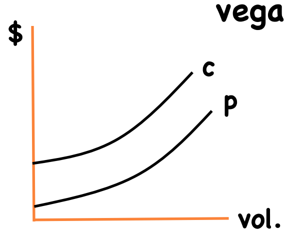
gamma/ - measures the risk that remains once a portfolio is delta neutral
- gamma risk can be managed to a specific level, but never eliminated



theta - a change in c or p for a change in time to expiration (rate of option decay)

long options - negative theta

short options - positive theta



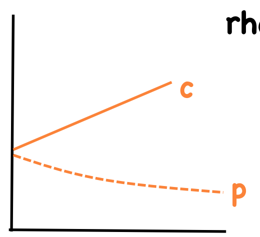
vega - change in c & p for a change in volatility

long options - positive vega

short options - negative vega

Review - 13

⇒ Option Greeks/



rho - a change in c & p for a change r_f

calls - pos. rho $\quad S_0e^{rT}$

puts - neg. rho $\quad Xe^{-rT}$

⇒ Implied Volatility/ - future volatility
- common to see different implied vols. for different exercise prices
  - volatility smile/skew